

TrES-2b

R-2270

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110525 · created 2026-08-16T22:52:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455706.82978 +/- 8.2e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1328 +/- 0.0077
Transit depth (Rp/Rs)^2	1.76 +/- 0.2 [%]
Orbital Inclination (inc)	84.29 +/- 0.3
Airmass coefficient 1 (a1)	1.2207 +/- 0.0066
Airmass coefficient 2 (a2)	-0.1824 +/- 0.0054
Scatter in the residuals of the lightcurve fit is	0.54 %
Best Comparison Star	None
Optimal Aperture	4.01
Optimal Annulus	11.66
Transit Duration (day)	0.0816 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1328 ± 0.0077 vs 0.1254 (deviation 0.96σ)
Duration fit vs published	0.0816 d vs 0.0746 d (deviation 1.67σ)
Mid-time O–C	0.0 min
Depth SNR	17.2
Residual scatter	0.54 %

Light curve

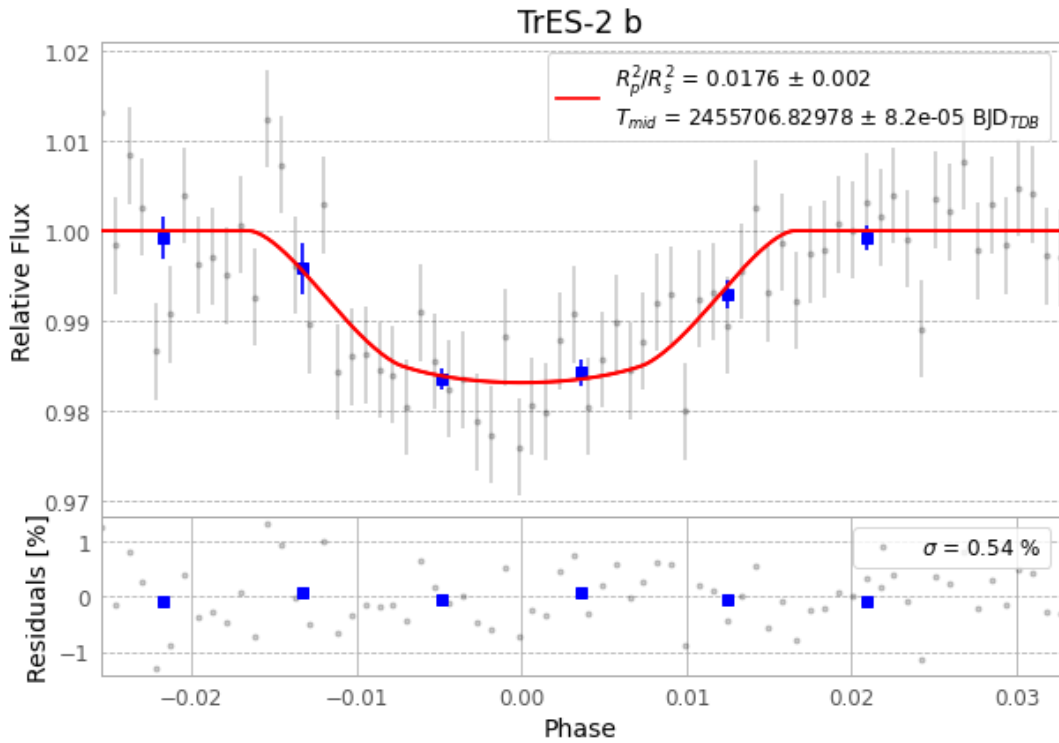


Figure 1 — FinalLightCurve_TrES-2 b_2011-05-24.png (SHA-256 9e6f04bba1a2feb1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4465caf4fb6209bf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), TRES-2110525062113.FITS → TRES-2110525094813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110525.log (63d16d908149...), FinalLightCurve_TrES-2 b_2011-05-24.png (9e6f04bba1a2...), FinalLightCurve_TrES-2 b_2011-05-24.csv (ff8b37d0df61...)

Compute resources

Wall time 107.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 22:52Z.